Meridian College of Medicine (fictional)

BMS 501

Foundations of Infection and Immunity

Case 2: Jordan Rivera

Self-Study Guide · Week 2

FICTIONAL SAMPLE for practice with the Med Student AI Study Agent. Invented school, faculty, and patient. Not affiliated with any medical school.

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Course Director Notes

Welcome to Case 2

This week follows Jordan Rivera, a college athlete with a painful skin infection. The case connects innate immunity, the virulence of Staphylococcus aureus, and how we choose antibiotics when resistance is possible.

Work through the case presentation first, then each discipline in order. Try every self-study question before you look at the answers.

Small-group session: Thursday, 9:00-11:00 (fictional).

This guide demonstrates AI-assisted study; follow your own school's AI policy for real course materials.

Case Presentation

Chief complaint

“There’s a painful bump on my arm that keeps getting bigger.”

History of present illness

Jordan Rivera is a 19-year-old college wrestler who presents to student health with a 4-day history of a red, swollen, painful lesion on the ulnar (medial) side of the right forearm. It began as what Jordan thought was a spider bite after practice. Over 48 hours it became firm and warm, and this morning it felt soft in the center. Jordan reports subjective fevers and fatigue. Two teammates had similar “boils” this season.

Past history

- No chronic illnesses; no prior hospitalizations
- Medications: none. Allergies: no known drug allergies
- Social: shares mats and towels at practice; no injection drug use

Vital signs

Measure	Value	Reference
Temperature	Value: 38.4 °C	Reference: 36.1-37.2 °C
Heart rate	Value: 104 /min	Reference: 60-100 /min
Blood pressure	Value: 118/72 mmHg	Reference: <120/80 mmHg
Respiratory rate	Value: 16 /min	Reference: 12-20 /min
SpO2	Value: 99% room air	Reference: ≥95%

Physical examination

A 4 cm erythematous, warm, tender nodule on the ulnar side of the right volar forearm with a fluctuant center and 2 cm of surrounding erythema. No lymphangitic streaking. Small, tender right epitrochlear node. The rest of the exam is normal.

Laboratory results

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Test	Result	Flag	Reference range	Units
WBC	Result: 14.2	Flag: H	Reference range: 4.0-11.0	Units: $\times 10^3/\mu L$
Neutrophils	Result: 82	Flag: H	Reference range: 40-70	Units: %
Band forms	Result: 6	Flag: H	Reference range: 0-5	Units: %
Lymphocytes	Result: 11	Flag: L	Reference range: 20-40	Units: %
Hemoglobin	Result: 14.6	Flag: none	Reference range: 13.5-17.5	Units: g/dL
Platelets	Result: 268	Flag: none	Reference range: 150-400	Units: $\times 10^3/\mu L$
C-reactive protein	Result: 48	Flag: H	Reference range: <10	Units: mg/L
Creatinine	Result: 0.9	Flag: none	Reference range: 0.7-1.3	Units: mg/dL
Glucose (random)	Result: 94	Flag: none	Reference range: 70-125	Units: mg/dL

Reference ranges are typical adult values (hemoglobin and creatinine shown for an adult male); they vary by laboratory.

Procedure and microbiology

Bedside incision and drainage released 6 mL of thick yellow pus. Gram stain showed many neutrophils and gram-positive cocci in clusters. Culture grew large golden colonies with beta-hemolysis on blood agar.

Test	Result	Interpretation
Catalase	Result: Positive	Interpretation: Staphylococcus
Coagulase	Result: Positive	Interpretation: S. aureus
Cefoxitin disk screen	Result: Resistant	Interpretation: mecA / PBP2a likely (MRSA)
Trimethoprim-sulfamethoxazole	Result: Susceptible	Interpretation: none
Doxycycline	Result: Susceptible	Interpretation: none
Clindamycin (D-test)	Result: Negative D-test; susceptible	Interpretation: none

Review the results above. Before reading on, write down: (1) the organism, (2) the host cells doing most of the work, and (3) what you would treat with and why.

Immunology

Discipline Director: Dr. Avery Lin (fictional)

avery.lin@meridian.example

Office hours: Tuesday 3:00-4:00 (fictional)

Discipline Director Notes:

- Focus on the first hours of infection: barriers, recognition, recruitment, and killing.
- Be able to trace a neutrophil from the bloodstream to the abscess.

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Self-Study Topics:

Videos:

Barriers and Pattern Recognition (LO-1101) (14:20)

Neutrophil Recruitment and Extravasation (LO-1102) (12:05)

Phagocytosis and Oxidative Killing (LO-1103) (10:42)

Slides:

Innate Immunity I: Barriers and Recognition (LO-1101)

Innate Immunity II: Neutrophils (LO-1102, LO-1103)

Notes: See Below

Learning Objectives:

1. Describe how skin acts as a physical, chemical, and microbial barrier.
2. Explain how pattern recognition receptors (TLR2) detect gram-positive bacteria.
3. Outline the steps of neutrophil recruitment: rolling, activation, adhesion, transmigration.
4. Explain how neutrophils kill bacteria with the respiratory burst.
5. Predict how S. aureus virulence factors interfere with each step.

Neutrophil Recruitment and Killing (LO-1102, LO-1103)

Recognition

- Bacterial lipoproteins are the main ligands for TLR2 (with TLR1 or TLR6); whether lipoteichoic acid also signals through TLR2 is debated [PMID:18406368].
- Tissue macrophages release IL-1, TNF-α, and IL-8 (CXCL8).

Recruitment (the leukocyte adhesion cascade)

Step	Endothelial molecule	Neutrophil molecule
Rolling	Endothelial molecule: E-selectin, P-selectin	Neutrophil molecule: Sialyl-Lewis X
Activation	Endothelial molecule: Chemokines (IL-8)	Neutrophil molecule: Chemokine receptors
Tight adhesion	Endothelial molecule: ICAM-1	Neutrophil molecule: Integrins (LFA-1, Mac-1)
Transmigration	Endothelial molecule: PECAM-1 (CD31)	Neutrophil molecule: PECAM-1 (CD31)

Killing

- Opsonized bacteria (IgG, C3b) are phagocytosed and the phagosome fuses with lysosomes.
- NADPH oxidase produces superoxide (the respiratory burst); myeloperoxidase makes hypochlorite.
- Pus is mostly dead neutrophils, bacteria, and tissue debris.

Clinical correlation

- Chronic granulomatous disease (NADPH oxidase defect): recurrent S. aureus, Aspergillus, and other infections. Often taught as 'catalase-positive organisms', but in a mouse model catalase-negative S. aureus was equally virulent [PMID:11997027].
- Leukocyte adhesion deficiency type 1 (CD18 defect): little or no pus with marked neutrophilia; delayed umbilical cord separation is classic but not universal.

Immunology: Self-Study Questions

1. A neutrophil rolling along the endothelium near Jordan’s abscess is slowed by which endothelial molecule?
    - A. ICAM-1

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- B. E-selectin
- C. PECAM-1
- D. VCAM-1
- E. MHC class II

2. A child has recurrent S. aureus abscesses and absent superoxide production on a nitroblue tetrazolium (NBT) test. Which enzyme is defective?

- A. Myeloperoxidase
- B. NADPH oxidase
- C. Superoxide dismutase
- D. Catalase
- E. Lysozyme

Immunology: Self-Study Questions + Answers

1. A neutrophil rolling along the endothelium near Jordan's abscess is slowed by which endothelial molecule?

- A. ICAM-1
- B. E-selectin
- C. PECAM-1
- D. VCAM-1
- E. MHC class II

Answer: B

Selectins mediate rolling; ICAM-1 binding to integrins mediates tight adhesion.

2. A child has recurrent S. aureus abscesses and absent superoxide production on a nitroblue tetrazolium (NBT) test. Which enzyme is defective?

- A. Myeloperoxidase
- B. NADPH oxidase
- C. Superoxide dismutase
- D. Catalase
- E. Lysozyme

Answer: B

No superoxide on NBT = absent respiratory burst = chronic granulomatous disease (NADPH oxidase defect). Myeloperoxidase deficiency spares superoxide production.

Microbiology

Discipline Director: Dr. Morgan Patel (fictional)

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Office hours: Wednesday 1:00-2:00 (fictional)

Discipline Director Notes:

- Know how to identify S. aureus in the lab and link each virulence factor to a disease.
- Practice the sample deck: sample/curriculum/MICRO/Staphylococcus aureus (sample lecture).pptx

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Self-Study Topics:

Videos:

Staphylococci: Identification (LO-1201) (9:40)

S. aureus Virulence Factors and Toxins (LO-1202) (16:15)

MRSA: Mechanism and Epidemiology (LO-1203) (8:30)

Slides:

Staphylococcus aureus (LO-1201, LO-1202, LO-1203)

Notes: See Below

Learning Objectives:

1. Identify S. aureus using Gram stain, catalase, and coagulase.
2. Match surface factors (protein A, capsule, coagulase) to how they evade the host.
3. Distinguish toxin-mediated diseases: TSS, scalded skin syndrome, food poisoning.
4. Explain the mechanism of methicillin resistance (mecA, PBP2a).
5. Contrast community-associated and hospital-associated MRSA.

Staphylococcus aureus (LO-1201, LO-1202, LO-1203)

Factor	Type	Action	Disease link
Protein A	Type: Surface	Action: Binds Fc of IgG; blocks opsonization	Disease link: Immune evasion
Coagulase	Type: Enzyme	Action: Activates prothrombin, forming fibrin	Disease link: Fibrin wall around the abscess
Panton-Valentine leukocidin	Type: Toxin	Action: Pore-forming; kills leukocytes	Disease link: Associated with CA-MRSA skin infection and necrotizing pneumonia (causal role debated)
TSST-1	Type: Superantigen	Action: Cross-links MHC II and TCR	Disease link: Toxic shock syndrome
Exfoliative toxin	Type: Protease	Action: Cleaves desmoglein-1	Disease link: Scalded skin syndrome
Enterotoxin	Type: Superantigen; heat-stable	Action: Preformed toxin in food	Disease link: Vomiting 30 min-8 h after eating (mean about 3 h)

Methicillin resistance

- mecA (on SCCmec) encodes PBP2a, a penicillin-binding protein with low affinity for beta-lactams.
- MRSA is resistant to nearly all beta-lactams (exception: anti-MRSA cephalosporins such as ceftaroline).
- Community-associated strains often carry PVL (its role in virulence is debated) and cause skin abscesses in healthy young people, especially with shared equipment and skin contact (athletes, dormitories) [PMID:17146447][PMID:22759601].

Microbiology: Self-Study Questions

1. Which test distinguishes S. aureus from coagulase-negative staphylococci?
    - A. Catalase
    - B. Coagulase
    - C. Optochin

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- D. Bacitracin
- E. Bile solubility

2. Jordan's isolate is cefoxitin-resistant. What is the molecular basis?

- A. Beta-lactamase hyperproduction
- B. Altered PBP (PBP2a) encoded by mecA
- C. Efflux pump
- D. D-Ala-D-Lac cell wall precursor
- E. Decreased porin expression

Microbiology: Self-Study Questions + Answers

1. Which test distinguishes S. aureus from coagulase-negative staphylococci?

- A. Catalase
- B. Coagulase
- C. Optochin
- D. Bacitracin
- E. Bile solubility

Answer: B

All staphylococci are catalase positive; among human isolates, S. aureus is the coagulase-positive species.

2. Jordan's isolate is cefoxitin-resistant. What is the molecular basis?

- A. Beta-lactamase hyperproduction
- B. Altered PBP (PBP2a) encoded by mecA
- C. Efflux pump
- D. D-Ala-D-Lac cell wall precursor
- E. Decreased porin expression

Answer: B

PBP2a binds beta-lactams poorly, so cell-wall synthesis continues.

Pharmacology

Discipline Director: Dr. Riley Okafor (fictional)

riley.okafor@meridian.example

Office hours: Friday 10:00-11:00 (fictional)

Discipline Director Notes:

- Know how each anti-staphylococcal drug works and when MRSA changes the choice.
- Incision and drainage is the main treatment for an abscess. Jordan has fever and tachycardia (two SIRS criteria), so an MRSA-active antibiotic is indicated [PMID:24973422]. In a large trial, TMP-SMX after drainage also raised the cure rate of uncomplicated abscesses (80.5% vs 73.6% with placebo, modified intention-to-treat) [PMID:26962903].

Summary of Important Symbols and Parameters

Symbol	Meaning	Formula
Vd	Meaning: Volume of distribution	Formula: Dose / C0
CL	Meaning: Clearance	Formula: Rate of elimination / plasma concentration
t½	Meaning: Half-life	Formula: 0.693 × Vd / CL
Css	Meaning: Steady-state concentration	Formula: Dosing rate / CL
LD	Meaning: Loading dose	Formula: Css × Vd / F

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Self-Study Topics:

Videos:

Cell Wall Inhibitors I: Beta-lactams (LO-1301) (13:10)

Cell Wall Inhibitors II: Vancomycin (LO-1302) (7:55)

Folate Antagonists and Tetracyclines (LO-1303) (11:20)

Slides:

Antibiotics for Gram-Positive Infections (LO-1301, LO-1302, LO-1303)

Notes: No Written Notes

Learning Objectives:

1. Explain how beta-lactams and vancomycin inhibit cell-wall synthesis, and why each fails against resistant organisms.
2. Choose an oral agent for uncomplicated CA-MRSA skin infection and justify it.
3. Name key adverse effects of vancomycin, TMP-SMX, and doxycycline.
4. Calculate a loading dose from Vd and a target concentration.

Drug	Target	Covers MRSA?	Key adverse effects
Nafcillin / oxacillin	Target: PBPs (beta-lactamase resistant)	Covers MRSA?: No	Key adverse effects: Neutropenia; interstitial nephritis (rare)
Cefazolin	Target: PBPs	Covers MRSA?: No	Key adverse effects: Hypersensitivity
Vancomycin	Target: Binds D-Ala-D-Ala	Covers MRSA?: Yes	Key adverse effects: Nephrotoxicity; infusion reaction (flushing)
TMP-SMX	Target: DHFR + dihydropteroate synthase	Covers MRSA?: Yes	Key adverse effects: Hyperkalemia, rash, bone marrow suppression
Doxycycline	Target: 30S ribosome	Covers MRSA?: Yes	Key adverse effects: Photosensitivity, esophagitis; avoid in pregnancy

Pharmacology: Self-Study Questions

1. After drainage, Jordan is sent home on an oral antibiotic. Which is most appropriate given the susceptibilities?
   - A. Dicloxacillin
   - B. Cephalexin
   - C. Trimethoprim-sulfamethoxazole
   - D. Amoxicillin
   - E. Penicillin V
2. A drug has Vd = 50 L and target concentration 20 mg/L with F = 1. What loading dose is needed?
   - A. 100 mg
   - B. 250 mg
   - C. 1,000 mg
   - D. 2,500 mg
   - E. 5,000 mg

Pharmacology: Self-Study Questions + Answers

1. After drainage, Jordan is sent home on an oral antibiotic. Which is most appropriate given the susceptibilities?

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- A. Dicloxacillin
- B. Cephalexin
- C. Trimethoprim-sulfamethoxazole
- D. Amoxicillin
- E. Penicillin V

Answer: C

The isolate is MRSA and TMP-SMX susceptible; beta-lactams do not cover MRSA.

2. A drug has Vd = 50 L and target concentration 20 mg/L with F = 1. What loading dose is needed?

- A. 100 mg
- B. 250 mg
- C. 1,000 mg
- D. 2,500 mg
- E. 5,000 mg

Answer: C

LD = Css × Vd / F = 20 mg/L × 50 L = 1,000 mg.

Selected References

Every reference below was checked in PubMed. A citation shows a source exists; open it to see whether it supports the claim.

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- Malachowa N, Kobayashi SD, DeLeo FR. Community-associated MRSA and athletes. Phys Sportsmed 2012. [PMID:22759601]
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- Zähringer U, et al. TLR2 - promiscuous or specific? Immunobiology 2008. [PMID:18406368]
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